TERM 3

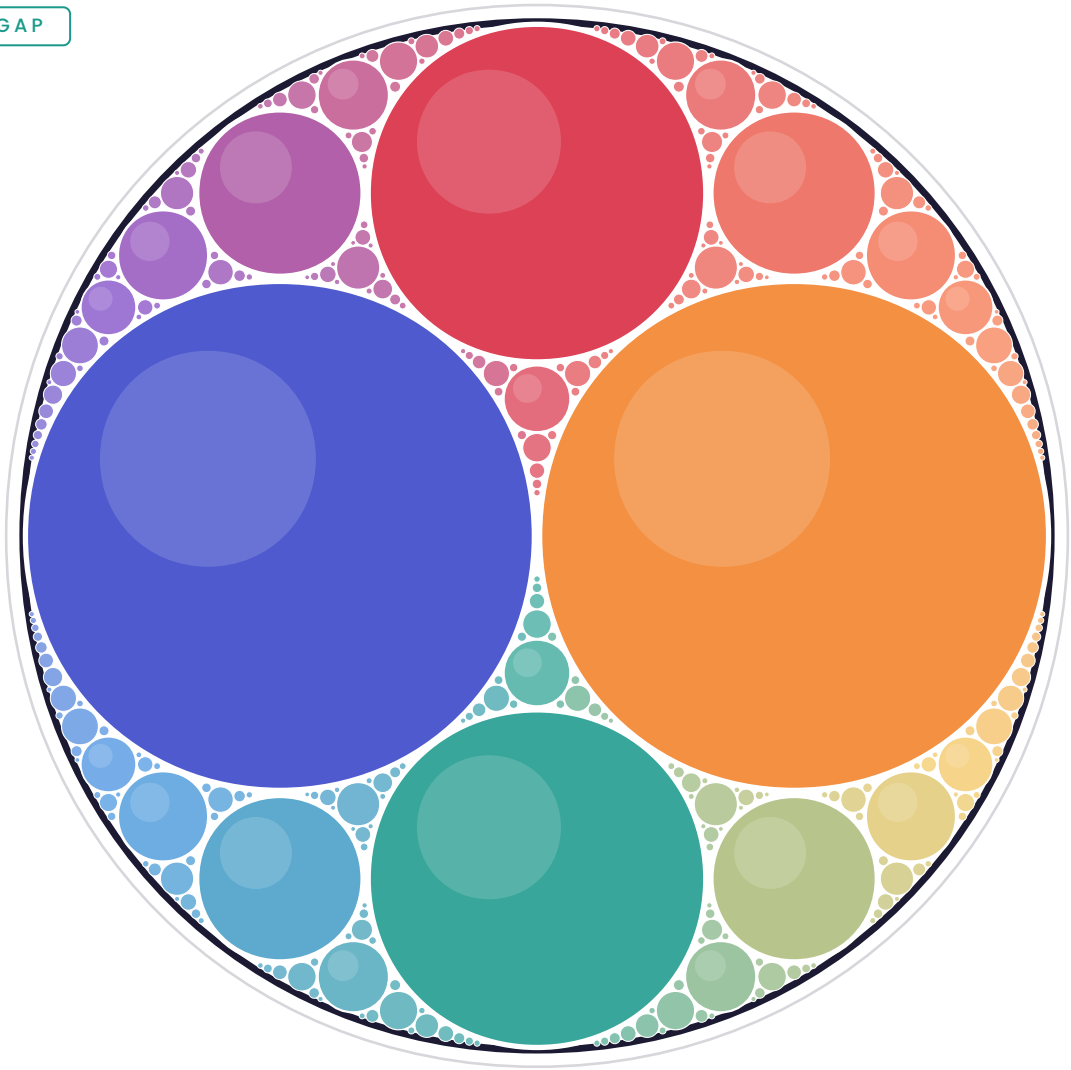
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FILL THE GAP



NAME _____

CLASS _____

APOLLONIAN GASKET

each circle fills a gap between three touching circles; every curvature (1 ÷ radius) is an integer

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Use the formula to find w when $P = 34$ and $l = 14$.

FORMULA $P = 2l + 2w$

COPY

SUBSTITUTE

ANSWER

2 Simplify.

$$5m^2 + 2m - 2m^2 + 5m$$

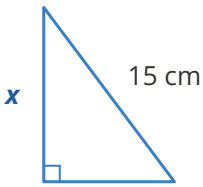
METHOD Like terms have exactly the same pronumerals and _____

WRITE IT IN FULL

WORKING

ANSWER

3 Find the length of the shorter side, x.



FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Evaluate.

$$10 - 12$$

METHOD Subtracting moves _____ on a number line

WRITE IT IN FULL

WORKING

ANSWER

5 The probability that the counter is red is $\frac{5}{8}$. Find the probability that the counter is not red.

FORMULA P(not E) = ___ − P(E)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Is $m = 3$ the solution of this equation? Show how you know. EXPLAIN

$$6m + 12 = m + 27$$

METHOD Check a solution by _____ it into both sides

WRITE IT IN FULL

WORKING

EXPLAIN

2 The price of a surf lesson changed from $50 to $70. Find the percentage increase.

FORMULA Percentage change = change ÷ _____ × 100%

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Find the value of this expression when $a = -3$ and $b = -5$.

$$3a - b$$

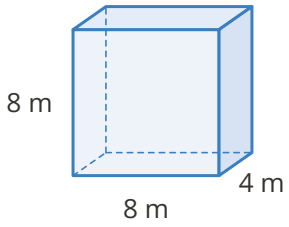
METHOD Replace each pronumeral with its _____ (use brackets)

WRITE IT IN FULL

WORKING

ANSWER

4 Find the volume of this rectangular prism.



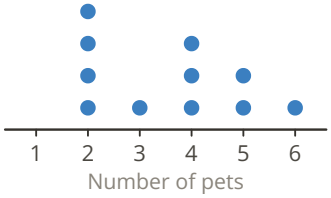
FORMULA $V = l \times w \times$ _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 The dot plot shows the number of pets each student in a class owns. Find the range.



FORMULA Range = _____ score − lowest score

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$m^2 = 4$$

METHOD $x^2 = c$ has ___ solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

WRITE IT IN FULL

WORKING

ANSWER

2 Expand.

$$5(5a + 7)$$

FORMULA $a(b + c) = ab +$ ___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 A farm paddock covers 6 ha. Convert this to m^2.

FORMULA 1 cm^2 = _____ mm^2, 1 m^2 = 10 000 cm^2, 1 ha = 10 000 m^2

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Heads came up 30 times in 50 tosses of a coin. Find the observed probability of heads.

FORMULA Observed probability = frequency ÷ _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Simplify. Write your answer as a single power.

$$9^5 \div 9^3$$

FORMULA $a^m \div a^n = a$___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Noah thinks of a number, multiplies it by 5 and subtracts 1. The answer is 9. Write and solve an equation to find the number.

METHOD Let n be the number; write an _____, then solve it

WRITE IT IN FULL

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$(4^4)^2$$

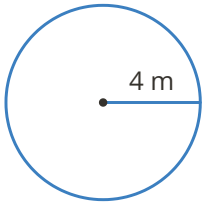
FORMULA $(a^m)^n = a$___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Find the circumference of this circle, correct to 1 decimal place.



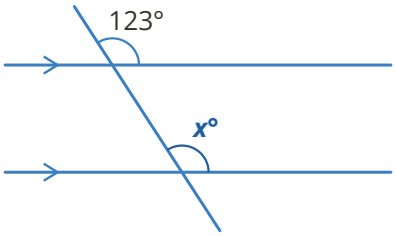
FORMULA $C = 2 \times$ _____ $\times r$

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Corresponding angles on parallel lines are _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Factorise.

$$7x - 7$$

METHOD Take out the highest common _____ (HCF); check by expanding

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$7n + 12 = 6n + 19$$

METHOD Move the _____ to one side first, then solve

WRITE IT IN FULL

WORKING

ANSWER

2 A bike helmet costs $60. It is on sale at 50% off. Find the sale price.

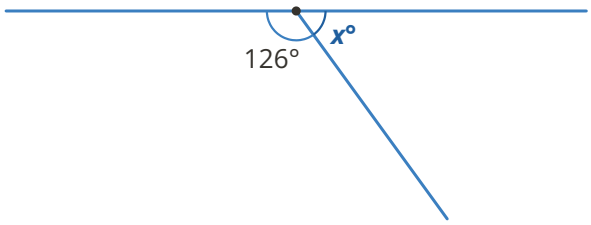
METHOD Discount = % × price; sale price = price ___ discount

WRITE IT IN FULL

WORKING

ANSWER

3 Find the value of x.



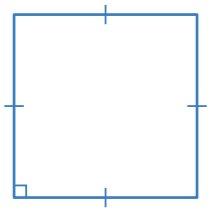
FORMULA Angles on a straight line add to _____°

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the perimeter of this square.



FORMULA $P =$ _____ $\times s$

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Simplify.

$$4m \times 5m$$

METHOD Multiply the numbers, then the pronumerals ($a \times a =$ ____)

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

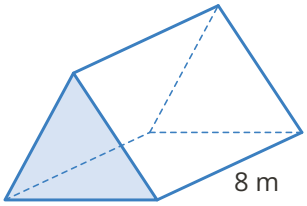
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5
1 The angle sum of a polygon with n sides is $S°$. Find n when $S = 1080$.	FORMULA $S = 180(n - 2)$ COPY SUBSTITUTE ANSWER	
2 Decrease 250 kg by 20%.	METHOD Find the percentage of the amount, then _____ it away WRITE IT IN FULL WORKING ANSWER	
3 Does the point (1, −3) lie on the line $y = x - 4$? Explain. EXPLAIN	METHOD A point lies on a line if its coordinates make the rule _____ WRITE IT IN FULL WORKING EXPLAIN	
4 Expand and simplify. $2(m + 6) + 8$	METHOD Expand the brackets, then collect _____ terms WRITE IT IN FULL WORKING ANSWER	
5 The shaded cross-section has an area of 6 m^2. Find the volume of this prism. 	FORMULA $V = A \times$ _____ WRITE IT IN FULL SUBSTITUTE ANSWER	

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve. Give your answer as a fraction or a decimal.

$$5m + 6 = 12$$

METHOD Do the same to _____ sides (use the inverse operation)

WRITE IT IN FULL

WORKING

ANSWER

2 Expand.

$$4(2k - 3)$$

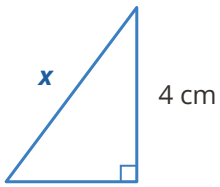
FORMULA $a(b + c) = ab +$ ___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Find the length of the hypotenuse, x.



FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the range of these scores.

Scores	11	9	16	18	6

FORMULA Range = _____ score − lowest score

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Mia says $3^4 = 12$. Explain her mistake and give the correct value. EXPLAIN

FORMULA $a^n = a \times a \times \ldots \times a$ (___ factors)

WRITE IT IN FULL

WORKING

EXPLAIN

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION | WORKING | SCORE /5

1 A market stall buys a candle for $10 and sells it for $14. Find the profit as a percentage of the cost price.

FORMULA Profit (or loss) % = profit (or loss) ÷ _____ × 100%

WRITE IT IN FULL

SUBSTITUTE

ANSWER

2 Simplify. Write your answer as a single power.

$$3^4 \times 3^5$$

FORMULA $a^m \times a^n = a$—

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Find the mean of these goal tallies.

Goals	16	10	5	9	15

FORMULA Mean = sum of scores ÷ _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Solve.

$$6k + 4 = 5k + 8$$

METHOD Move the _____ to one side first, then solve

WRITE IT IN FULL

WORKING

ANSWER

5 Find the area of this rectangle.



3 cm

9 cm

FORMULA $A = l \times$ ____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$m^2 + 3 = 39$$

METHOD $x^2 = c$ has ___ solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

WRITE IT IN FULL

WORKING

ANSWER

2 Factorise.

$$10y - 25$$

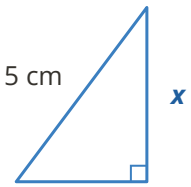
METHOD Take out the highest common _____ (HCF); check by expanding

WRITE IT IN FULL

WORKING

ANSWER

3 Find the length of the shorter side, x.



FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Calculate.

$$\frac{3}{8} \text{ of } 80 \text{ L}$$

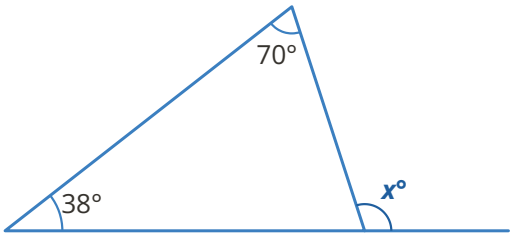
FORMULA Fraction of a quantity = fraction ___ quantity

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Exterior angle = _____ of the 2 interior opposite angles

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The angle sum of a polygon with n sides is $S°$. Find n when $S = 360$.

FORMULA $S = 180(n - 2)$

COPY

SUBSTITUTE

ANSWER

2 Increase 100 mL by 5%.

METHOD Find the percentage of the amount, then _____ it on

WRITE IT IN FULL

WORKING

ANSWER

3 Write 200 as a product of prime factors in index form.

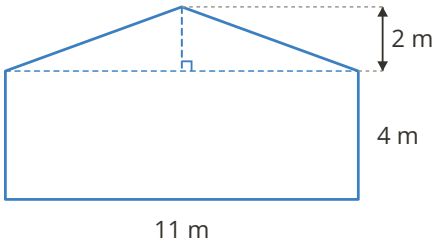
METHOD Factor tree to primes, then write repeated primes as _____

WRITE IT IN FULL

WORKING

ANSWER

4 Find the area of this shape.



METHOD Split the shape into rectangles (and triangles), then _____ their areas

WRITE IT IN FULL

WORKING

ANSWER

5 A letter is chosen at random from this word. Find P(C).

S U C C E S S

FORMULA P(E) = number of favourable outcomes ÷ _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Mia expanded $6(p + 8)$ and got $6p + 8$. Explain the error. EXPLAIN

FORMULA $a(b + c) = ab + $ ___

WRITE IT IN FULL

WORKING

EXPLAIN

2 Evaluate.

$$6 \times 10^0$$

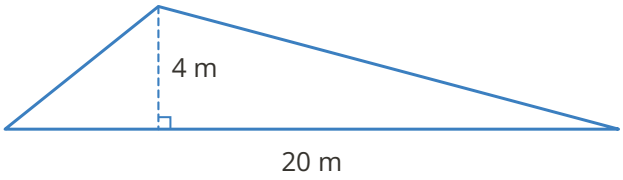
FORMULA $a^0 =$ ___ (any non-zero number to the power 0)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Find the area of this triangle.



FORMULA $A =$ ____ $\times b \times h$

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Felix thinks of a number, multiplies it by 5 and subtracts 12. The answer is 3. Write and solve an equation to find the number.

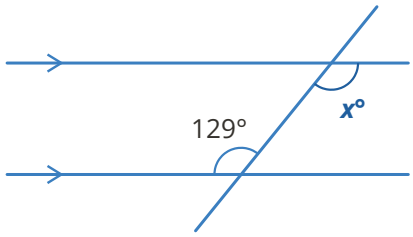
METHOD Let n be the number; write an _____, then solve it

WRITE IT IN FULL

WORKING

ANSWER

5 Find the value of x.



FORMULA Alternate angles on parallel lines are _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand and simplify.

$$3(n + 2) + 2$$

METHOD Expand the brackets, then collect ____ terms

WRITE IT IN FULL

WORKING

ANSWER

2 Solve.

$$4k + 12 = k + 30$$

METHOD Move the _____ to one side first, then solve

WRITE IT IN FULL

WORKING

ANSWER

3 A triangle has sides 15 cm, 20 cm and 25 cm. Is it right-angled?

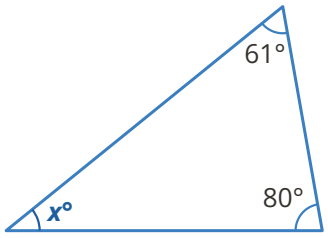
FORMULA Right-angled if (longest side)^2 = _____ of the squares of the other 2 sides

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Angle sum of a triangle = ____°

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Calculate.

40% of $60

FORMULA Percentage of a quantity = percentage ÷ ____ × quantity

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$3^8 \div 3^4$$

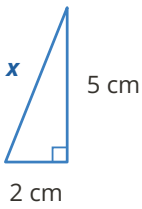
FORMULA $a^m \div a^n = a$___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

2 Find the hypotenuse, x, correct to 1 decimal place.



FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Solve.

$$2y^2 = 98$$

METHOD $x^2 = c$ has ___ solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

WRITE IT IN FULL

WORKING

ANSWER

4 Write as a recurring decimal (use dot notation).

$$\frac{4}{9}$$

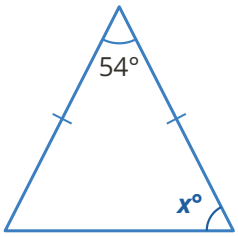
METHOD Numerator ÷ denominator; put a ___ over each repeating digit

WRITE IT IN FULL

WORKING

ANSWER

5 Find the value of x.



FORMULA Isosceles triangle: base angles are _____; angle sum = 180°

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Factorise.

$$24y + 64$$

METHOD Take out the highest common _____ (HCF); check by expanding

WRITE IT IN FULL

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$(5^4)^2$$

FORMULA $(a^m)^n = a$___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Find the median of these heights.

cm	149	144	158	152	153	137	139

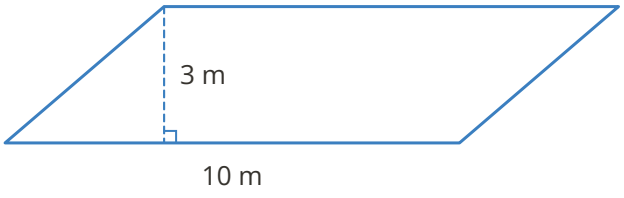
FORMULA Median = middle score when the scores are in _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the area of this parallelogram.



FORMULA $A = b \times$ ____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Hiring a kayak at Cudgen Creek costs C dollars for h hours. Use the formula to find h when $C = 92$.

FORMULA $C = 12h + 20$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify.

$$6m^2 + 3m - 2m^2 + 4m$$

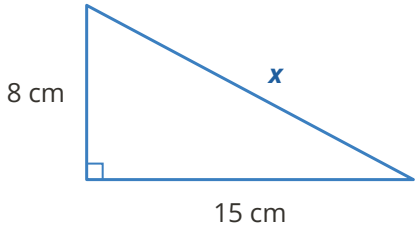
METHOD Like terms have exactly the same pronumerals and _____

WRITE IT IN FULL

WORKING

ANSWER

2 Find the length of the hypotenuse, x.



FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 The Kingscliff touch footy team won 27 of its 30 games. What percentage of its games did it win?

FORMULA Percentage = part ÷ whole × _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Solve.

$$7n + 4 = n + 16$$

METHOD Move the _____ to one side first, then solve

WRITE IT IN FULL

WORKING

ANSWER

5 Which has the larger area: a 10 cm by 5 cm rectangle or a square with sides of 7 cm? Explain. EXPLAIN

FORMULA $A = lw$ and $A = s$—

WRITE IT IN FULL

WORKING

EXPLAIN

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve. Give your answer as a fraction or a decimal.

$$5a + 1 = 4$$

METHOD Do the same to _____ sides (use the inverse operation)

WRITE IT IN FULL

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$5^6 \times 5^3$$

FORMULA $a^m \times a^n = a$___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Expand.

$$2(5a - 1)$$

FORMULA $a(b + c) = ab +$ ___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 A bag holds 6 red and 7 white marbles. One is chosen at random. Is it more likely than not to be red? Explain. EXPLAIN

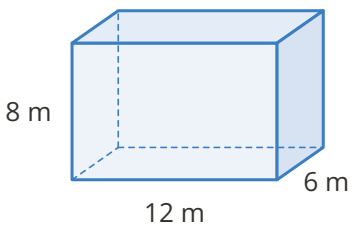
FORMULA P(E) = number of favourable outcomes ÷ _____

WRITE IT IN FULL

WORKING

EXPLAIN

5 Find the volume of this rectangular prism.



FORMULA $V = l \times w \times$ ____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

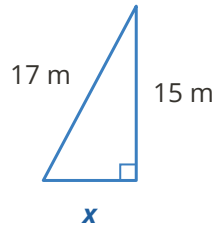
TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the shorter side, x.



FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

2 Write 56 as a product of prime factors in index form.

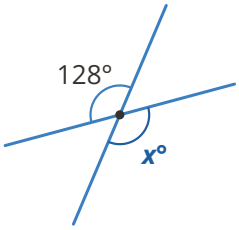
METHOD Factor tree to primes, then write repeated primes as _____

WRITE IT IN FULL

WORKING

ANSWER

3 Find the value of x.



FORMULA Vertically opposite angles are _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the mode of these scores.

Scores	5	7	9	1	7	2	2	6	7

FORMULA Mode = the most _____ score

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Solve.

$$3m^2 = 12$$

METHOD $x^2 = c$ has ___ solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the value of this expression when $a = -3$ and $b = 5$.

$$3a - b$$

METHOD Replace each pronumeral with its _____ (use brackets)

WRITE IT IN FULL

WORKING

ANSWER

2 40% of a number is 64. What is the number?

METHOD Unitary method: find _____ (or 10%) first, then multiply to get 100%

WRITE IT IN FULL

WORKING

ANSWER

3 Simplify.

$$30mn \div 5m$$

METHOD Divide the numbers, then _____ pronumerals that are top and bottom

WRITE IT IN FULL

WORKING

ANSWER

4 Aisha thinks of a number, doubles it and adds 6. The answer is 16. Write and solve an equation to find the number.

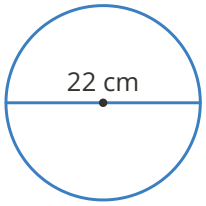
METHOD Let n be the number; write an _____, then solve it

WRITE IT IN FULL

WORKING

ANSWER

5 Find the circumference of this circle, correct to 1 decimal place.



FORMULA $C =$ _____ $\times d$

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Between which two whole numbers does this square root lie?

$$\sqrt{45}$$

METHOD Find the _____ numbers just below and just above the number

WRITE IT IN FULL

WORKING

ANSWER

2 Factorise.

$$9y + 3$$

METHOD Take out the highest common _____ (HCF); check by expanding

WRITE IT IN FULL

WORKING

ANSWER

3 Calculate.

$$0.038 \div 100$$

METHOD × 10, 100, 1000: digits move 1, 2, 3 places _____. ÷: they move right

WRITE IT IN FULL

WORKING

ANSWER

4 Solve.

$$5n + 3 = 3n + 7$$

METHOD Move the _____ to one side first, then solve

WRITE IT IN FULL

WORKING

ANSWER

5 Find the value of this expression when $p = 1$.

$$p^2 + 4$$

METHOD Replace each pronumeral with its _____ (use brackets)

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 A triangle has sides 5 cm, 6 cm and 8 cm. Is it right-angled?

FORMULA Right-angled if (longest side)^2 = ____ of the squares of the other 2 sides

WRITE IT IN FULL

SUBSTITUTE

ANSWER

2 Expand and simplify.

$$5(x + 2) + 6$$

METHOD Expand the brackets, then collect ____ terms

WRITE IT IN FULL

WORKING

ANSWER

3 Write as a decimal.

$$120\%$$

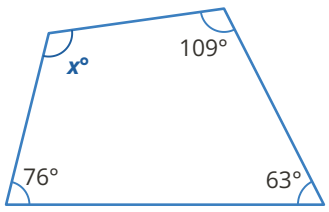
METHOD Percentage to decimal: percentage ÷ ____

WRITE IT IN FULL

WORKING

ANSWER

4 Find the value of x. Give a reason. REASON



FORMULA Angle sum of a quadrilateral = ____°

WRITE IT IN FULL

SUBSTITUTE

REASON

ANSWER

5 Use the formula to find t when $v = 7$, $u = 1$ and $a = 2$.

FORMULA $v = u + at$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify.

$$8p^2 \div 2p$$

METHOD Divide the numbers, then ____ common pronumerals ($a^2 \div a = a$)

WRITE IT IN FULL

WORKING

ANSWER

2 Solve.

$$5m^2 = 180$$

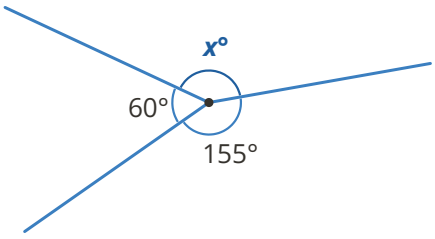
METHOD $x^2 = c$ has ___ solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

WRITE IT IN FULL

WORKING

ANSWER

3 Find the value of x.



FORMULA Angles at a point add to ____°

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 The price of a boogie board is $130 before GST. How much is the GST (10%)?

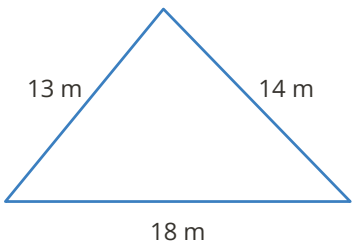
FORMULA GST = ____ of the price; price incl. GST = price + GST

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Find the perimeter of this triangle.



METHOD Perimeter = the _____ of all the side lengths

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand.

$$2(4k - 7)$$

FORMULA $a(b + c) = ab + $ ___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

2 The price of a pie changed from \$8 to \$10. Find the percentage increase.

FORMULA Percentage change = change ÷ _____ × 100%

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 The probability that the counter is red is 0.2. Find the probability that the counter is not red.

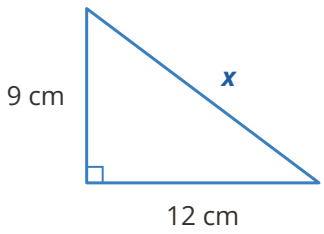
FORMULA P(not E) = ___ − P(E)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the length of the hypotenuse, x.



FORMULA $c^2 = a^2 + $ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Evaluate.

$$\sqrt[3]{125}$$

METHOD $\sqrt{a}$ is the number that _____ to a; $\sqrt[3]{a}$ is the number that cubes to a

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$7k + 7 = 4k + 19$$

METHOD Move the _____ to one side first, then solve

WRITE IT IN FULL

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$9^5 \div 9^2$$

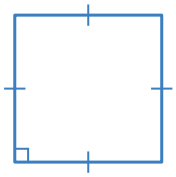
FORMULA $a^m \div a^n = a$—

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Find the area of this square.



4 cm

FORMULA $A = s$—

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 A pair of sneakers costs $200. It is on sale at 30% off. Find the sale price.

METHOD Discount = % × price; sale price = price ___ discount

WRITE IT IN FULL

WORKING

ANSWER

5 Heads came up 50 times in 100 tosses of a coin. Find the observed probability of heads.

FORMULA Observed probability = frequency ÷ _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Is 50% of \$60 the same as 60% of \$50? Explain. EXPLAIN

METHOD Percentage of a quantity = percentage ÷ ____ × quantity

WRITE IT IN FULL

WORKING

EXPLAIN

2 Factorise.

$$6y - 2$$

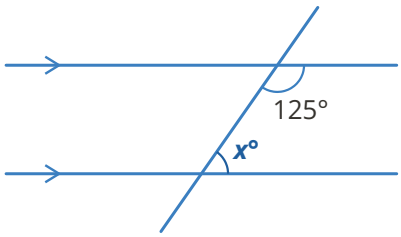
METHOD Take out the highest common _____ (HCF); check by expanding

WRITE IT IN FULL

WORKING

ANSWER

3 Find the value of x.



FORMULA Co-interior angles on parallel lines add to _____°

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Convert 750 cm^3 to millilitres.

FORMULA 1 cm^3 = 1 mL, 1 L = _____ mL, 1 m^3 = 1000 L

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Solve. Give your answer as a fraction or a decimal.

$$5m + 5 = 18$$

METHOD Do the same to _____ sides (use the inverse operation)

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

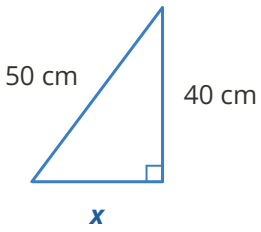
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5
1 Use the formula to find w when $P = 18$ and $l = 6$.	FORMULA $P = 2l + 2w$ COPY SUBSTITUTE ANSWER	
2 Expand and simplify. $3(a + 2) + 5a$	METHOD Expand the brackets, then collect _____ terms WRITE IT IN FULL WORKING ANSWER	
3 Find the length of the shorter side, x. 	FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse) WRITE IT IN FULL SUBSTITUTE ANSWER	
4 Is $k = 10$ the solution of this equation? Show how you know. EXPLAIN $5k + 12 = 67$	METHOD Check a solution by _____ it into both sides WRITE IT IN FULL WORKING EXPLAIN	
5 Calculate. $12.5 + 4.75$	METHOD Line up the decimal _____, then add WRITE IT IN FULL WORKING ANSWER	

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION | WORKING | SCORE /5

1 Expand.

$$2(2k + 2)$$

FORMULA $a(b + c) = ab + $ ___

WRITE IT IN FULL

SUBSTITUTE

ANSWER

2 Simplify. Write your answer as a single power.

$$(3^3)^4$$

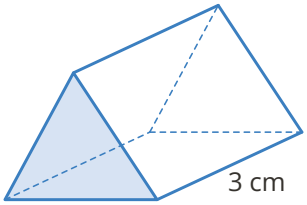
FORMULA $(a^m)^n = a$ —

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 The shaded cross-section has an area of 24 cm^2. Find the volume of this prism.



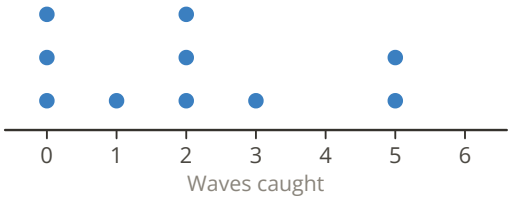
FORMULA $V = A \times$ ____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 The dot plot shows the waves each nipper caught in a session. Find the range.



FORMULA Range = _____ score − lowest score

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Decrease 20 L by 15%.

METHOD Find the percentage of the amount, then _____ it away

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

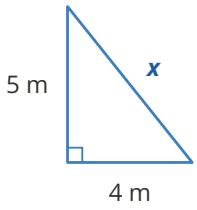
TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the hypotenuse, x, correct to 1 decimal place.



FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

2 A surf shop buys a T-shirt for $40 and sells it for $50. Find the profit as a percentage of the cost price.

FORMULA Profit (or loss) % = profit (or loss) ÷ _____ × 100%

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Solve.

$$5n + 7 = 4n + 16$$

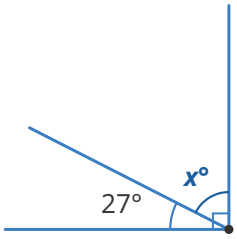
METHOD Move the _____ to one side first, then solve

WRITE IT IN FULL

WORKING

ANSWER

4 Find the value of x.



FORMULA Complementary angles add to _____°

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Solve.

$$k + 13 = 22$$

METHOD Do the same to _____ sides (use the inverse operation)

WRITE IT IN FULL

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$8^5 \times 8^4$$

FORMULA $a^m \times a^n = a^{__}$

WRITE IT IN FULL

SUBSTITUTE

ANSWER

2 Solve.

$$y^2 + 7 = 56$$

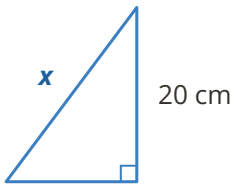
METHOD $x^2 = c$ has ___ solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

WRITE IT IN FULL

WORKING

ANSWER

3 Find the length of the hypotenuse, x.



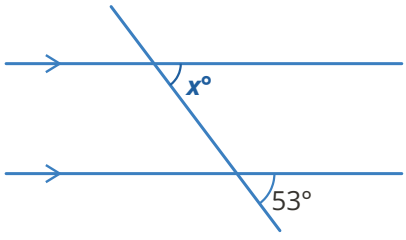
FORMULA $c^2 = a^2 +$ _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Corresponding angles on parallel lines are _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 The stem-and-leaf plot shows the ages of people at a beach. Find the median.

1	5 7
2	
3	0 8 9

Key: 1 | 5 = 15

FORMULA Median = middle score when the scores are in _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5
1 Increase $250 by 40%.	METHOD Find the percentage of the amount, then ____ it on WRITE IT IN FULL WORKING ANSWER	
2 A triangle has sides 4 m, 6 m and 7 m. Is it right-angled?	FORMULA Right-angled if (longest side)^2 = ____ of the squares of the other 2 sides WRITE IT IN FULL SUBSTITUTE ANSWER	
3 Priya thinks of a number, doubles it and adds 9. The answer is 27. Write and solve an equation to find the number.	METHOD Let n be the number; write an _____, then solve it WRITE IT IN FULL WORKING ANSWER	
4 Simplify. $7 \times 2m$	METHOD Multiply the _____, then the pronumerals (in alphabetical order) WRITE IT IN FULL WORKING ANSWER	
5 Sam says 6 + 11 × 9 = 153. Explain the mistake and give the correct answer. EXPLAIN	METHOD Brackets, then × and ÷ (left to right), then _____ (left to right) WRITE IT IN FULL WORKING EXPLAIN	

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 3 Fill the gap in each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Use the formula to find w when $P = 30$ and $l = 13$.

FORMULA $P = 2l + 2w$

COPY

SUBSTITUTE

ANSWER

2 Evaluate.

$$3^0 + 4^0$$

FORMULA a^0 = ___ (any non-zero number to the power 0)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

3 Find the mean of these waves caught by nippers.

Waves	0	2	2	3	1	4

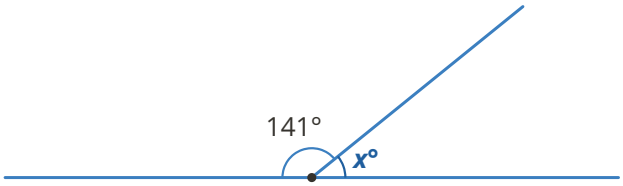
FORMULA Mean = sum of scores ÷ _____

WRITE IT IN FULL

SUBSTITUTE

ANSWER

4 Find the value of x.



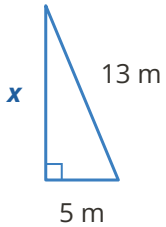
FORMULA Angles on a straight line add to _____°

WRITE IT IN FULL

SUBSTITUTE

ANSWER

5 Find the length of the shorter side, x.



FORMULA $c^2 = a^2$ + _____ (c is the hypotenuse)

WRITE IT IN FULL

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Formula · Reference 1

TERM 3 Every formula and rule used in this booklet.

NUMBER

FRACTIONS AND PERCENTAGES OF A QUANTITY

Fraction of a quantity = fraction × quantity

Percentage of a quantity = percentage ÷ 100 × quantity

Percentage = part ÷ whole × 100%

POWERS AND ROOTS

$a^n = a \times a \times \ldots \times a$ (n factors)

$\sqrt{a}$ is the number that squares to a; $\sqrt[3]{a}$ is the number that cubes to a

CONVERTING FRACTIONS, DECIMALS AND PERCENTAGES

Percentage to decimal: percentage ÷ 100

INTEGERS

Subtracting moves left on a number line

MULTIPLYING AND DIVIDING BY 10, 100, 1000

× 10, 100, 1000: digits move 1, 2, 3 places left. ÷: they move right

OPERATIONS WITH DECIMALS

Line up the decimal points, then add

ORDER OF OPERATIONS

Brackets, then × and ÷ (left to right), then + and − (left to right)

RECURRING DECIMALS

Numerator ÷ denominator; put a dot over each repeating digit

INDEX LAWS · YEAR 8

$a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$

$a^m \times a^n = a^{m+n}$

$a^0 = 1$ (any non-zero number to the power 0 is 1)

PERCENTAGE INCREASE AND DECREASE · YEAR 8

Percentage change = change ÷ original amount × 100%

Discount = % × price; sale price = price − discount

Find the percentage of the amount, then take it away

Find the percentage of the amount, then add it on

PROFIT, LOSS AND GST · YEAR 8

Profit (or loss) % = profit (or loss) ÷ cost price × 100%

GST = 10% of the price; price incl. GST = price + GST

PRIME FACTORS IN INDEX FORM · YEAR 8

Factor tree to primes, then write repeated primes as powers

THE UNITARY METHOD · YEAR 8

Unitary method: find 1% (or 10%) first, then multiply to get 100%

ALGEBRA AND EQUATIONS

MULTIPLYING AND DIVIDING TERMS

Multiply the numbers, then the pronumerals ($a \times a = a^2$)

Divide the numbers, then cancel pronumerals that are top and bottom

Divide the numbers, then cancel common pronumerals ($a^2 \div a = a$)

Multiply the numbers, then the pronumerals (in alphabetical order)

SOLVING EQUATIONS

Check a solution by substituting it into both sides

Do the same to both sides (use the inverse operation)

LINEAR RELATIONSHIPS

A point lies on a line if its coordinates make the rule true

SUBSTITUTION

Replace each pronumeral with its value (use brackets)

EQUATION PROBLEMS · YEAR 8

Let n be the number; write an equation, then solve it

EQUATIONS OF THE FORM $AX^2 = C$ · YEAR 8

$x^2 = c$ has 2 solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

EXPANDING AND SIMPLIFYING · YEAR 8

Expand the brackets, then collect like terms

EXPANDING BRACKETS · YEAR 8

$a(b + c) = ab + ac$

FACTORISING · YEAR 8

Take out the highest common factor (HCF); check by expanding

LIKE TERMS · YEAR 8

Like terms have exactly the same pronumerals and powers

PRONUMERALS ON BOTH SIDES · YEAR 8

Move the pronumerals to one side first, then solve

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Formula · Reference 2

TERM 3 Every formula and rule used in this booklet (continued).

MEASUREMENT

AREA OF RECTANGLES AND SQUARES

$A = lw$

$A = lw$ and $A = s^2$

$A = s^2$

CIRCUMFERENCE OF A CIRCLE

$C = 2\pi r$

$C = \pi d$

PERIMETER

$P = 4s$

Perimeter = the sum of all the side lengths

UNITS OF AREA, VOLUME AND CAPACITY

1 cm^2 = 100 mm^2, 1 m^2 = 10 000 cm^2, 1 ha = 10 000 m^2

1 cm^3 = 1 mL, 1 L = 1000 mL, 1 m^3 = 1000 L

VOLUME OF PRISMS

$V = lwh$

$V = Ah$

AREA OF A COMPOSITE SHAPE

Split the shape into rectangles (and triangles), then add their areas

AREA OF A PARALLELOGRAM

$A = bh$

AREA OF A TRIANGLE

$A = \frac{1}{2}bh$

PYTHAGORAS' THEOREM · YEAR 8

$c^2 = a^2 + b^2$ (c is the hypotenuse)

Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

Find the square numbers just below and just above the number

GEOMETRY: ANGLES

ANGLES AT A POINT AND ON A LINE

Angles on a straight line add to 180°

Vertically opposite angles are equal

Angles at a point add to 360°

Complementary angles add to 90°

TRIANGLES AND QUADRILATERALS

Exterior angle = sum of the 2 interior opposite angles

Angle sum of a triangle = 180°

Isosceles triangle: base angles are equal; angle sum = 180°

Angle sum of a quadrilateral = 360°

ANGLES ON PARALLEL LINES

Corresponding angles on parallel lines are equal

Alternate angles on parallel lines are equal

Co-interior angles on parallel lines add to 180°

STATISTICS AND PROBABILITY

MEAN, MEDIAN, MODE AND RANGE

Range = highest score − lowest score

Mean = sum of scores ÷ number of scores

Median = middle score when the scores are in order (2 middles: halfway)

Mode = the most common score

Median = middle score when the scores are in order

PROBABILITY

P(not E) = 1 − P(E)

Observed probability = frequency ÷ number of trials

P(E) = number of favourable outcomes ÷ total number of outcomes

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

My · Tracker

TERM 3 Write your score after each session.

NAME _____

DATE _____

WEEK	SESSION 1	SESSION 2	SESSION 3	WEEK TOTAL
Week 1	/ 5	/ 5	/ 5	/ 15
Week 2	/ 5	/ 5	/ 5	/ 15
Week 3	/ 5	/ 5	/ 5	/ 15
Week 4	/ 5	/ 5	/ 5	/ 15
Week 5	/ 5	/ 5	/ 5	/ 15
Week 6	/ 5	/ 5	/ 5	/ 15
Week 7	/ 5	/ 5	/ 5	/ 15
Week 8	/ 5	/ 5	/ 5	/ 15
Week 9	/ 5	/ 5	/ 5	/ 15
Week 10	/ 5	/ 5	/ 5	/ 15
TERM TOTAL				**/ 150**

OUTCOMES IN THIS BOOKLET · NSW MATHEMATICS K–10 (2022), STAGE 4

MA4-INT-C-01	compares, orders and calculates with integers to solve problems	NCM 7
MA4-FRC-C-01	represents and operates with fractions, decimals and percentages to solve problems	NCM 7, NCM 8
MA4-IND-C-01	operates with primes and roots, positive-integer and zero indices involving numerical bases and establishes the relevant index laws	NCM 7, NCM 8
MA4-ALG-C-01	generalises number properties to operate with algebraic expressions including expansion and factorisation	NCM 7, NCM 8
MA4-EQU-C-01	solves linear equations of up to 2 steps and quadratic equations of the form $ax^2 = c$	NCM 7, NCM 8
MA4-LIN-C-01	creates and displays number patterns and finds graphical solutions to problems involving linear relationships	NCM 7
MA4-LEN-C-01	applies knowledge of the perimeter of plane shapes and the circumference of circles to solve problems	NCM 7
MA4-ARE-C-01	applies knowledge of area and composite area involving triangles, quadrilaterals and circles to solve problems	NCM 7
MA4-VOL-C-01	applies knowledge of volume and capacity to solve problems involving right prisms and cylinders	NCM 7
MA4-PYT-C-01	applies Pythagoras' theorem to solve problems in various contexts	NCM 8
MA4-ANG-C-01	applies angle relationships to solve problems, including those related to transversals on sets of parallel lines	NCM 7
MA4-GEO-C-01	identifies and applies the properties of triangles and quadrilaterals to solve problems	NCM 7
MA4-DAT-C-02	analyses simple datasets using measures of centre, range and shape of the data	NCM 7
MA4-PRO-C-01	solves problems involving the probabilities of simple chance experiments	NCM 7
MAO-WM-01	Working mathematically: the weekly reasoning questions (marked EXPLAIN or REASON) ask students to explain, check or justify.	

Year 8 units in these warm-ups (each starts once its teaching ends in the 2027 Year 8 mixed ability program): Rational numbers, percentages & financial maths from T1 Wk 5; Index laws from T1 Wk 7; Pythagoras' theorem from T1 Wk 10; Algebra 2 from T2 Wk 4; Equations 2 from T2 Wk 9. All other questions revise Year 7 (NCM 7 Ch 1–11). Ratios, rates and time (NCM 7 Ch 12) are held back until Year 8 teaches them.